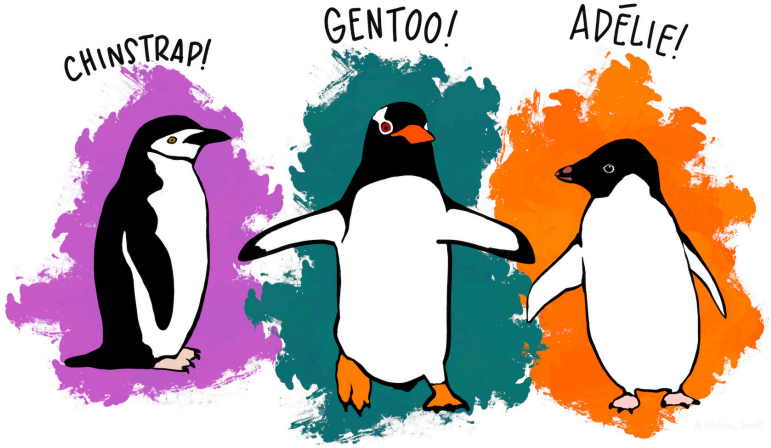
CHAPTER 5 · INTERPRETABLE MACHINE LEARNING

Data and Models

The datasets and models we interpret throughout the book

Asasira Martin

Source: C. Molnar, Interpretable Machine Learning, 3rd ed. (2025)



Palmer penguins · artwork by Allison Horst

and bike rentals in Washington, D.C.

OVERVIEW

Two datasets, two tasks

Bike rentals

REGRESSION

728 days

Target: number of bikes rented per day (cnt)

Capital Bikeshare, Washington, D.C.; weather and season info added by Fanaee-T and Gama (2014)

Palmer penguins

CLASSIFICATION

333 penguins

Target: sex (male / female)

Palmer Station, Antarctica; Gorman, Williams and Fraser (2014)

01

Bike rentals

Regression: predicting the number of bikes rented per day

BIKE RENTALS

One target, eight features

`cnt` TARGET
Count of rented bikes (casual + registered)

`season`
Spring, summer, fall or winter

`holiday`
Is the day a holiday?

`workday`
Working day or weekend?

`weather`
Weather: good, misty or bad

`temp`
Temperature (°C)

`hum`
Relative humidity (%, 0–100)

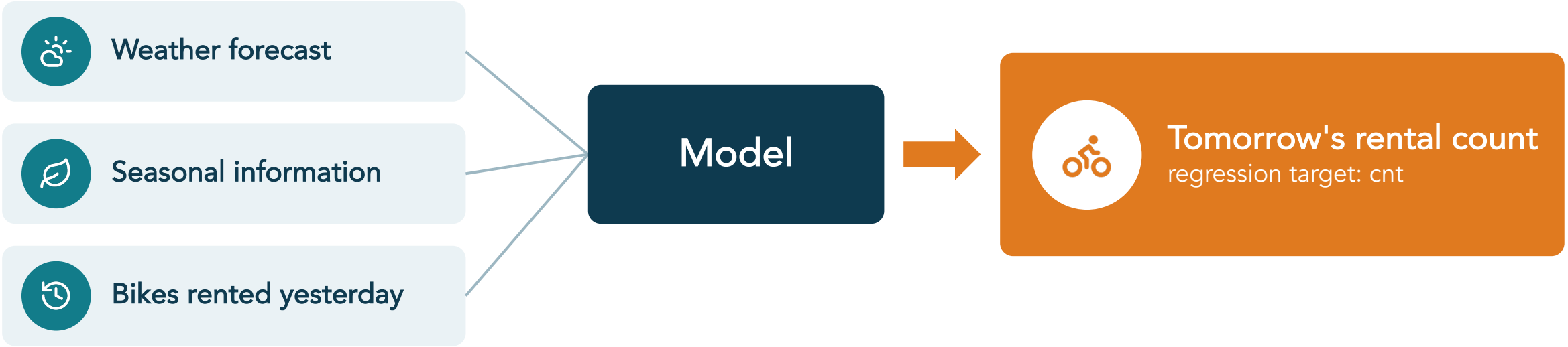
`windspeed`
Wind speed (km/h)

`cnt_2d_bfr`
Bikes rented two days before

Processing: one day with humidity 0 and the first two days (no count from two days before) were removed → 728 days.

BIKE RENTALS

The task: how many bikes will be rented tomorrow?



Simplifying assumption

The weather features are treated as forecasts; in reality they are measured values.

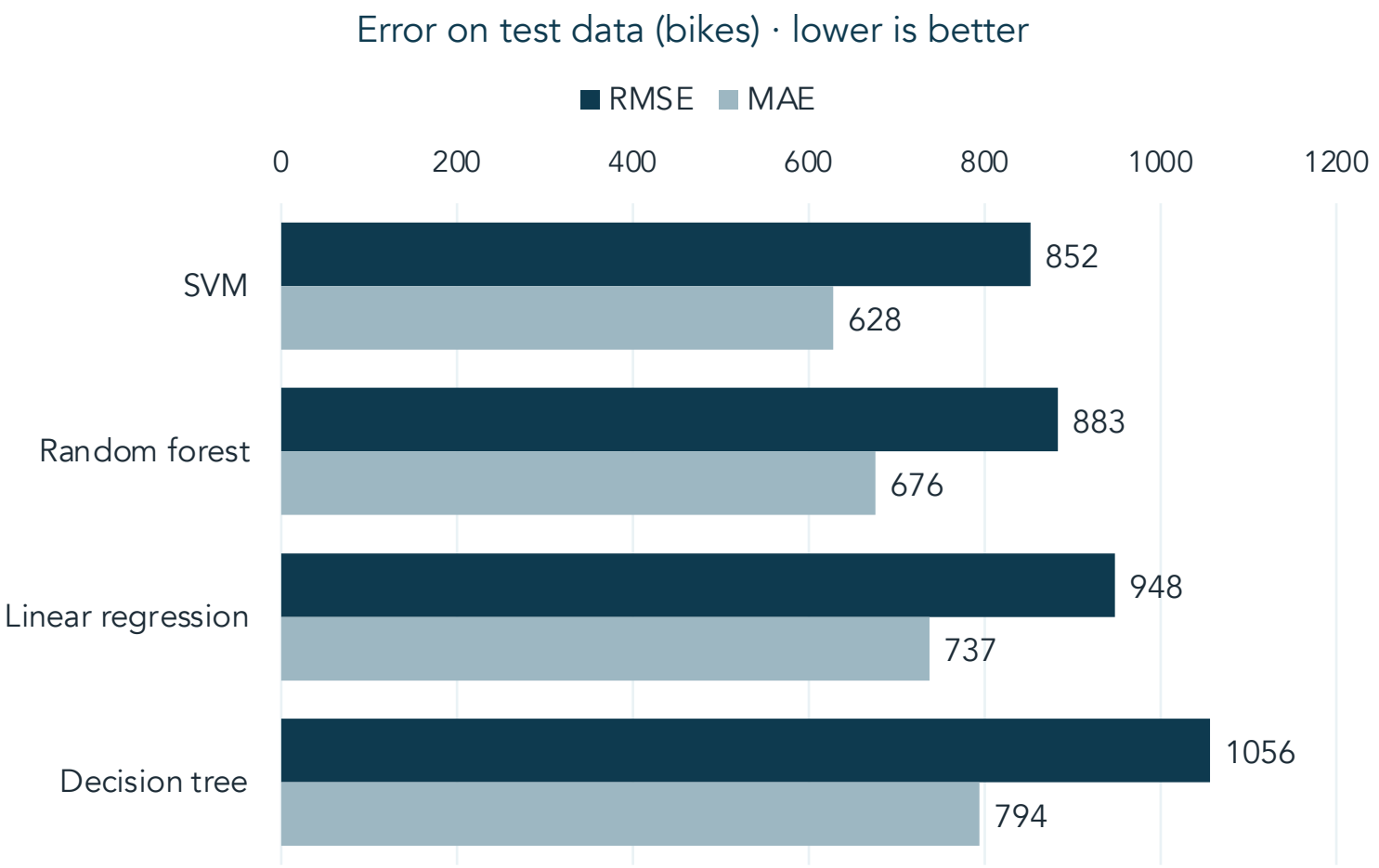
Holdout split



Four models

- Random forest
- CART tree
- SVM
- Linear regression

The SVM wins; the decision tree falls clearly behind



852

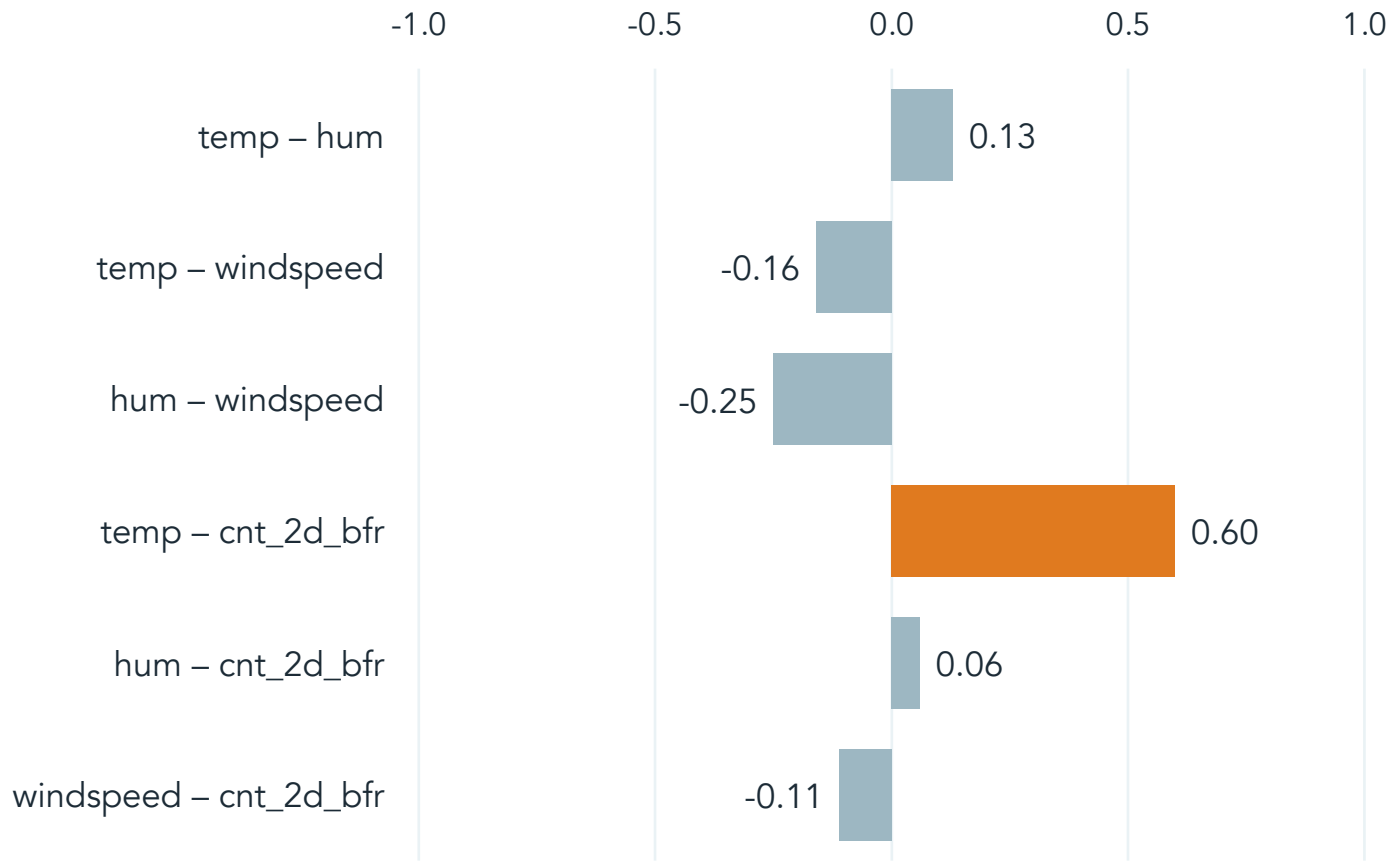
RMSE of the SVM: the lowest error on test data (MAE 628).

RMSE root mean squared error: punishes large errors more.

MAE mean absolute error: the average deviation.

One notable correlation: temperature and past rentals

Pearson correlation between the numerical features



But what about…
categorical features? non-linear relationships?

Two more tools

Pairwise scatter plots
Visualize the raw dependence

Normalized mutual information
Measures non-linear and categorical dependence

METHOD NOTE

Normalized mutual information (NMI)

Mutual information

$$MI(X_j, X_k) = \sum_{c,d} P(c,d) \cdot \log \left[P(c,d) / (P(c) \cdot P(d)) \right]$$

Normalized

$$NMI(X_j, X_k) = 2 \cdot MI(X_j, X_k) / \left[H(X_j) + H(X_k) \right]$$

Entropy

$$H(X_j) = - \sum_{c} P(c) \cdot \log P(c)$$

Numerical features: number of bins

$$nbins = \left[(max - min) / (2 \cdot IQR / n^{1/3}) \right] \quad \text{(Freedman–Diaconis)}$$

How to read it



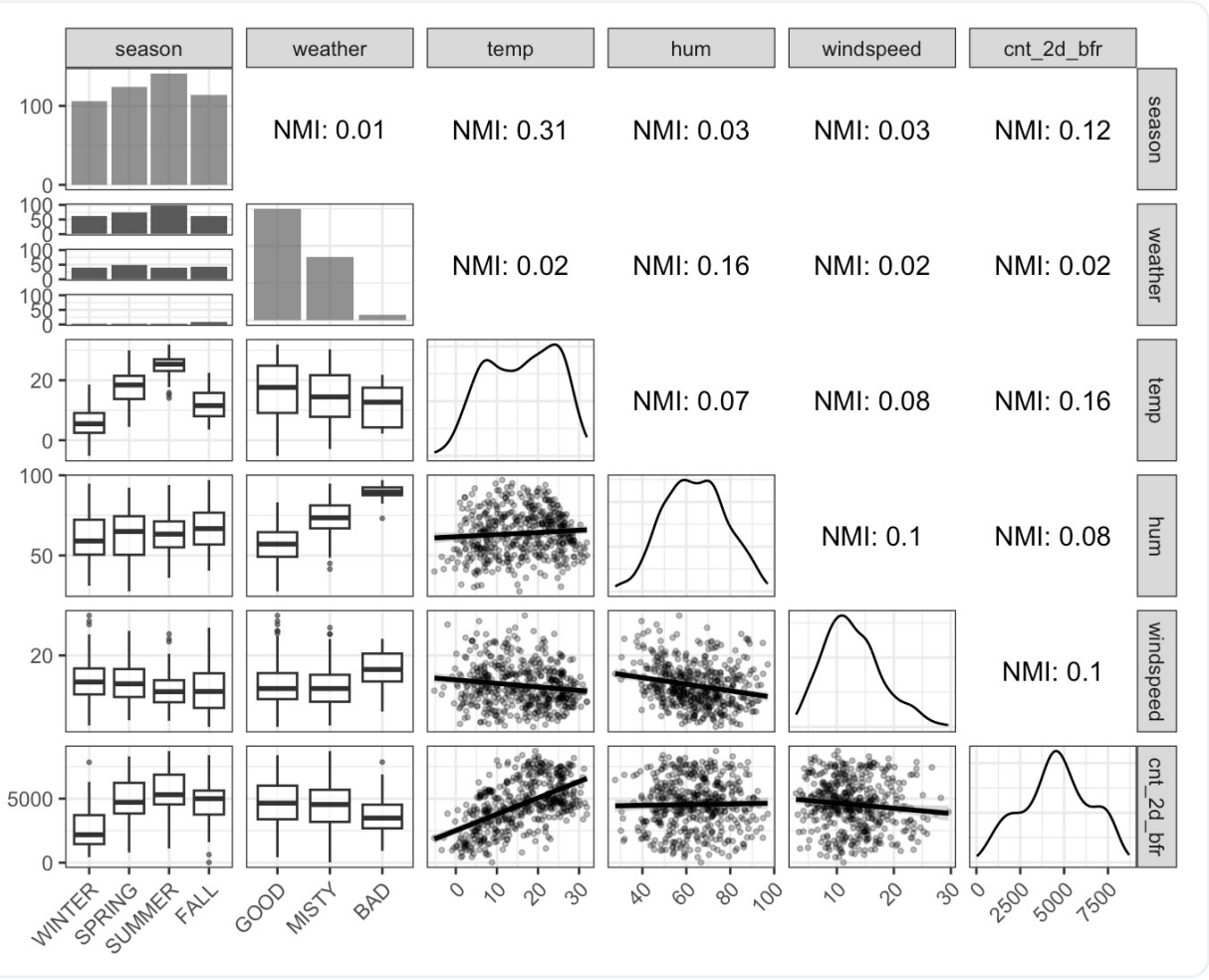
0
no shared information

1
all variation from the dependence

Caution

NMI can be biased upward for features with many bins or categories (Mahmoudi, 2024). So we read it together with raw-data plots and the Pearson correlation.

NMI confirms the correlations and covers categoricals



Same picture

NMI broadly confirms the impression from the correlation analysis.

Categorical features

Season shares information with temperature and with the count from two days before, which is no surprise.

Mostly linear

The pairwise plots show no unusual patterns, so correlation is a good measure here.

Figure: Molnar (2025). holiday and workday removed for readability.

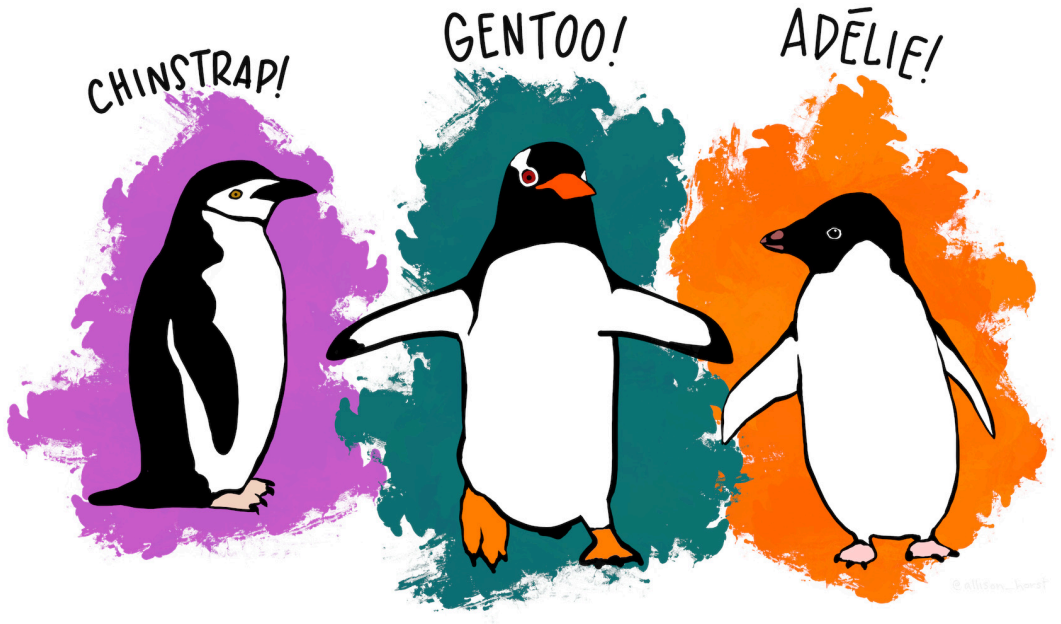
02

Palmer penguins

Classification: predicting a penguin's sex from body measurements

PALMER PENGUINS

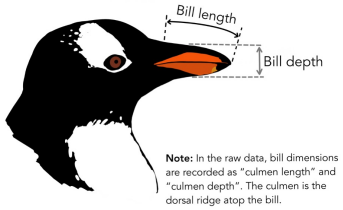
333 penguins, five features, one target



Three species: Chinstrap, Gentoo, Adelie · artwork by Allison Horst

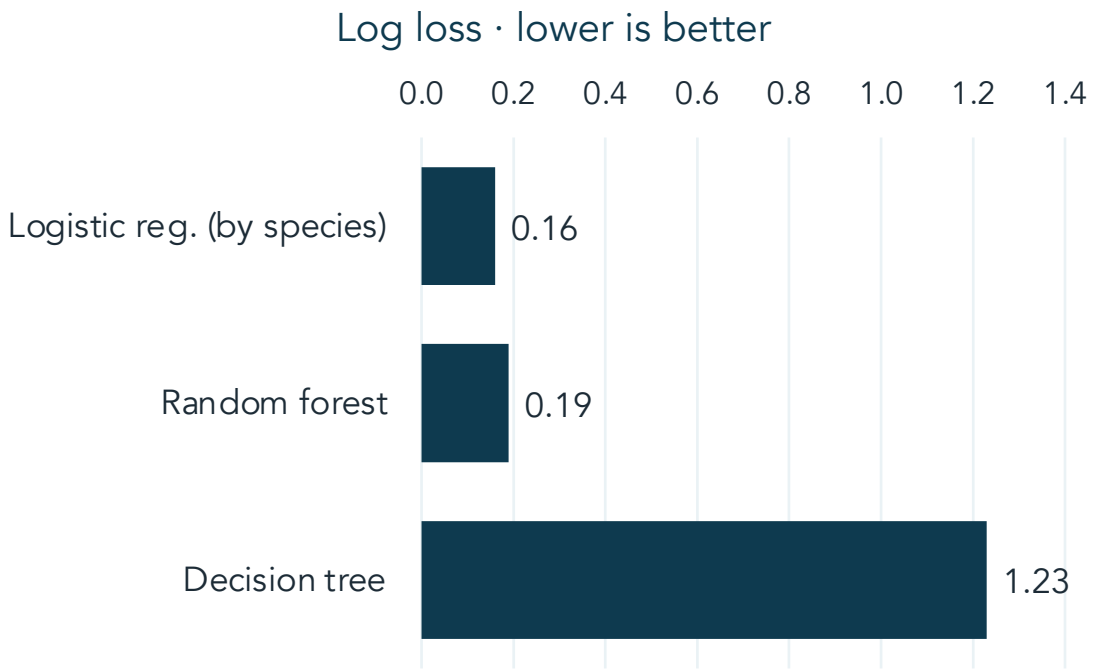
Source: Gorman, Williams and Fraser (2014), Palmer Station (LTER). 11 penguins with missing values were removed; loaded with the palmerpenguins R package.

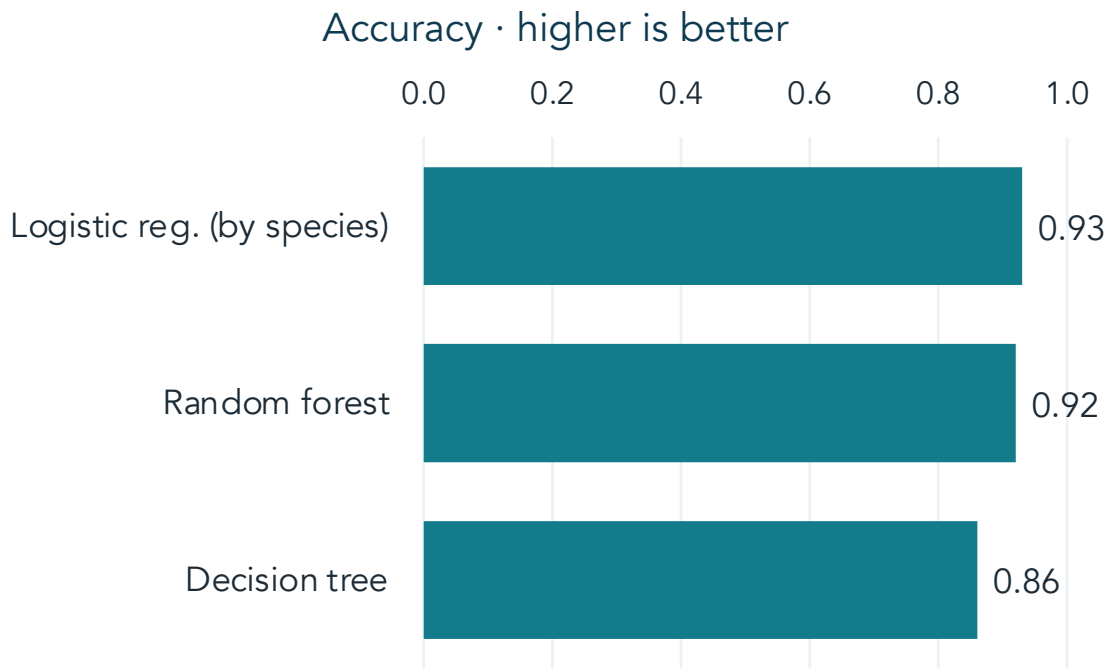
`sex`	**Male / female · TARGET**
`species`	Chinstrap, Gentoo or Adelie
`body_mass_g`	Body mass (g)
`bill_length_mm`	Bill length (mm)
`bill_depth_mm`	Bill depth (mm)
`flipper_length_mm`	Flipper length (mm)



How bill length and depth are measured. The study looks at differences in how female and male penguins look.

Logistic regression by species performs best



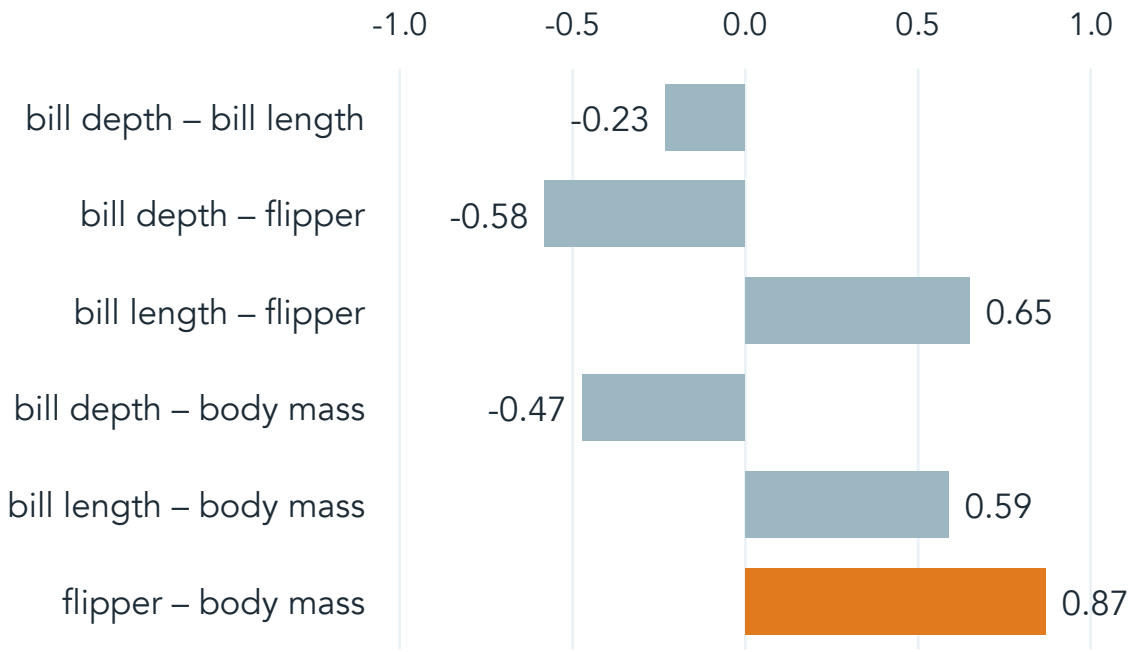


Logistic regression = 3 models
The data is split by species and one model is trained per species, as in Gorman et al. (2014).

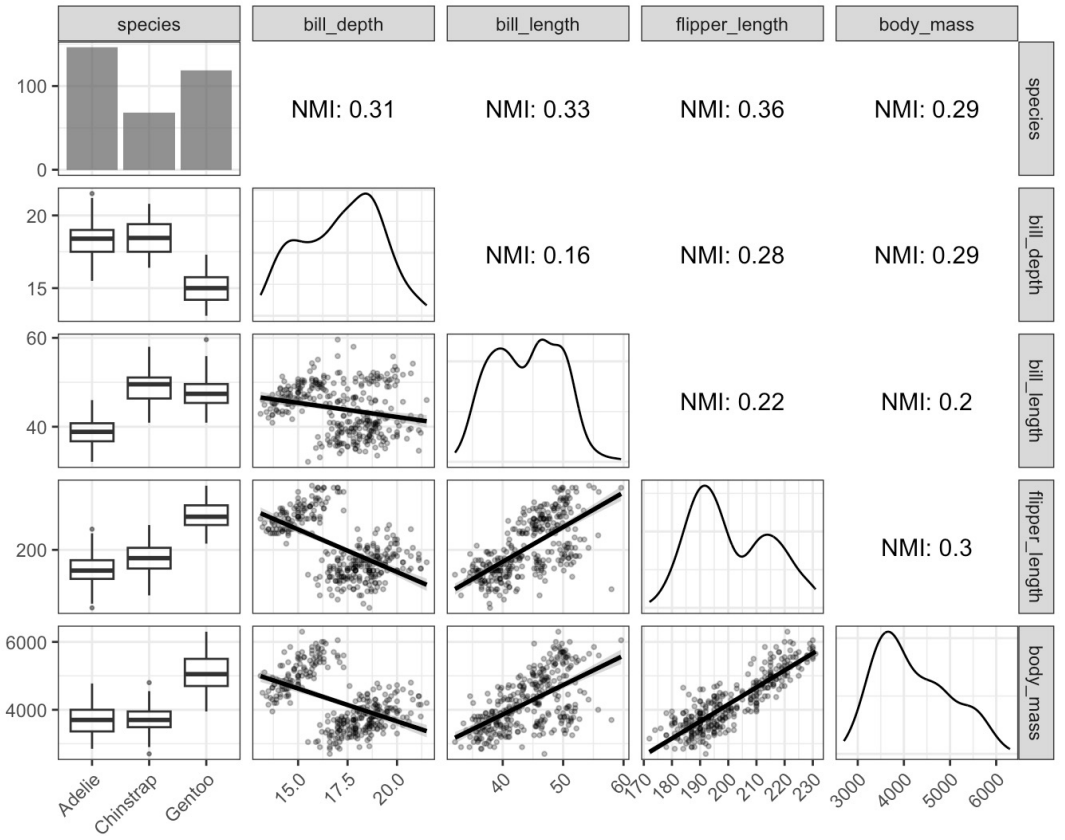
Trees use species as a feature
This didn't work well for the decision tree; the random forest gets close in accuracy.

Pearson correlation tells only half the story

Pearson correlation



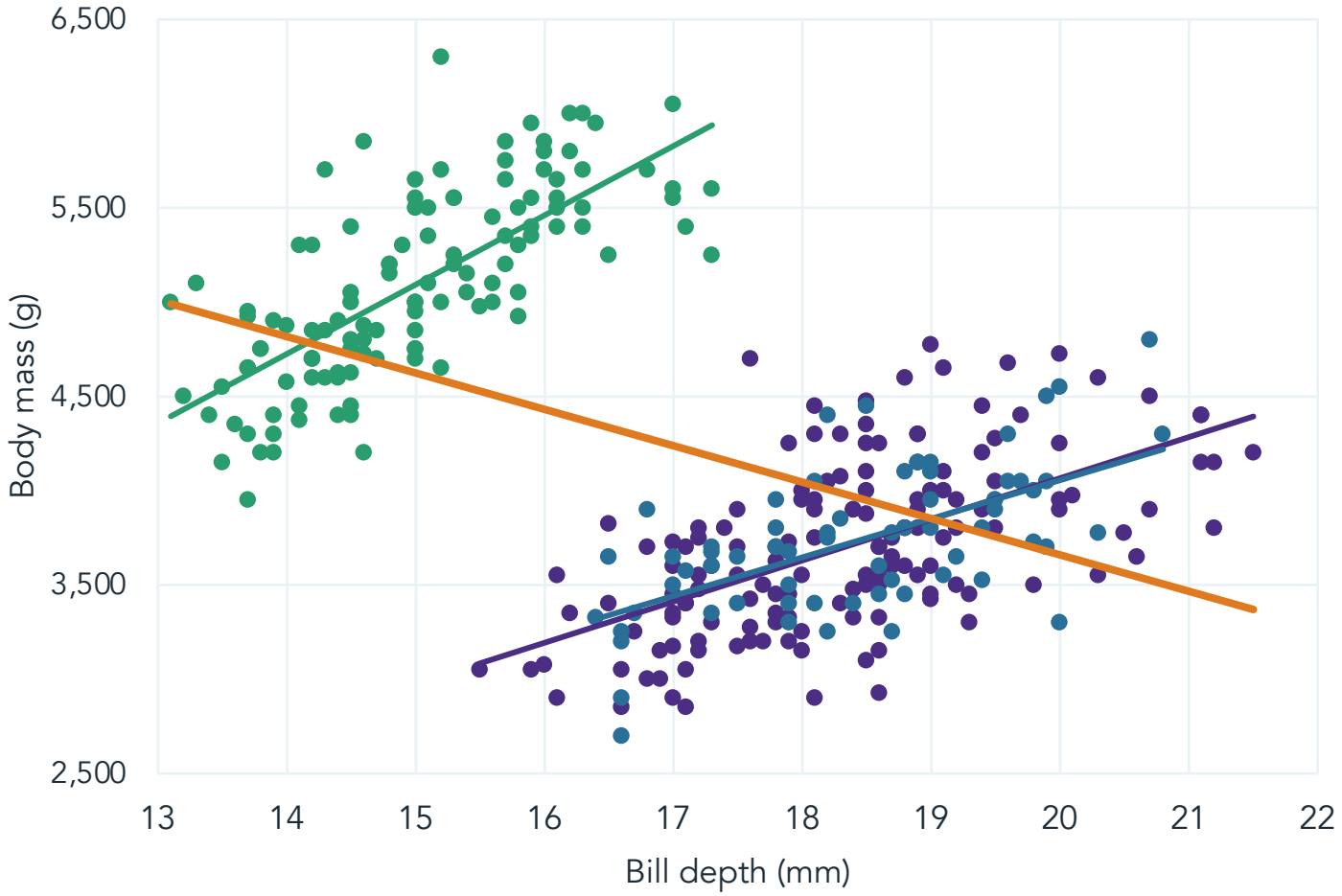
Notice: the NMI of body mass and bill depth is as high as body mass and flipper, but the correlation is much lower (−0.47).



Pairwise plots and NMI · Figure: Molnar (2025)

PALMER PENGUINS · SIMPSON'S PARADOX

Positive within each species, negative when pooled



Correlation of body mass and bill depth

Adelie	+0.58
Chinstrap	+0.60
Gentoo	+0.72
All penguins	**−0.47**

Why? Gentoo penguins are heavier but have shallower bills. Pooling the groups reverses the trend. Mutual information handles this.

A MODELING DECISION

One model, or separate models per group?

The book's default

All penguins in one dataset; species is a feature.

One model that learns from all the data.

or

The alternative

Model and analyze each species separately.

Arguably better, and logistic regression by species already won.

A very common tension in machine learning: data points belong to a cluster or entity



Predicting sales for different stores



Classifying lab samples that come from the same batches

SUMMARY

Know your data before you interpret

How features depend on each other directly shapes the results of interpretation methods.

Thank you · Questions?

Molnar, C. (2025). “Data and Models.” Interpretable Machine Learning, 3rd ed.



Bike rentals

Regression, 728 days. Best model: SVM (RMSE 852).



Palmer penguins

Classification, 333 penguins. Logistic regression by species (accuracy 0.93).



Dependence

Use correlation, NMI and plots together. Watch out for Simpson's paradox.